

# Opportunistic Target Selection

## Early Directional Commitment for Query-Efficient Black-Box Adversarial Attacks

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### 1. Motivation: Realistic Black-Box Attacks

#### Why black-box?

- Real attackers rarely know the model weights, architecture, or gradients.
- Many deployed systems still expose scores, confidence values, or ranked labels.
- Query-efficient score-based attacks are therefore the closest setting to a practical threat model.

Score-based black-box attacks see probabilities, not gradients. Standard untargeted losses mainly lower the true-class confidence:

$$\mathcal{L}_{\text{untarg}}(x', y) = P(y \mid x') \quad \text{or} \quad \log P(y \mid x').$$

- Probability mass is freed, but no non-true class is chosen.
- The trajectory can drift across several class basins.
- Query budget is wasted before misclassification.

**Goal.** Pick a viable target from the trajectory, then commit early to avoid drift.

#### Gap.

- Targeted attacks need a target before the run starts.
- Untargeted CE/probability losses do not decide which competitor should take over.

### 2. Method: Opportunistic Target Selection

OTS keeps the proposal mechanism unchanged. Only the loss schedule changes after a short untargeted prefix.

#### Algorithm.

1. Run the original untargeted attack for  $T$  iterations.
2. Lock onto the leading non-true class

$$t = \arg \max_{k \neq y} P(k \mid x'_T).$$

3. Continue the same attack, but maximize the target confidence  $P(t \mid x')$ .
4. Stop when the classifier prediction is no longer  $y$ .

- Irreversible lock: no repeated exploration.
- Target choice matters: random targets fail; the leading class after a short prefix is viable.
- Stable for  $T \in \{5, \dots, 500\}$ : rankings appear early.

### 3. Experimental Protocol

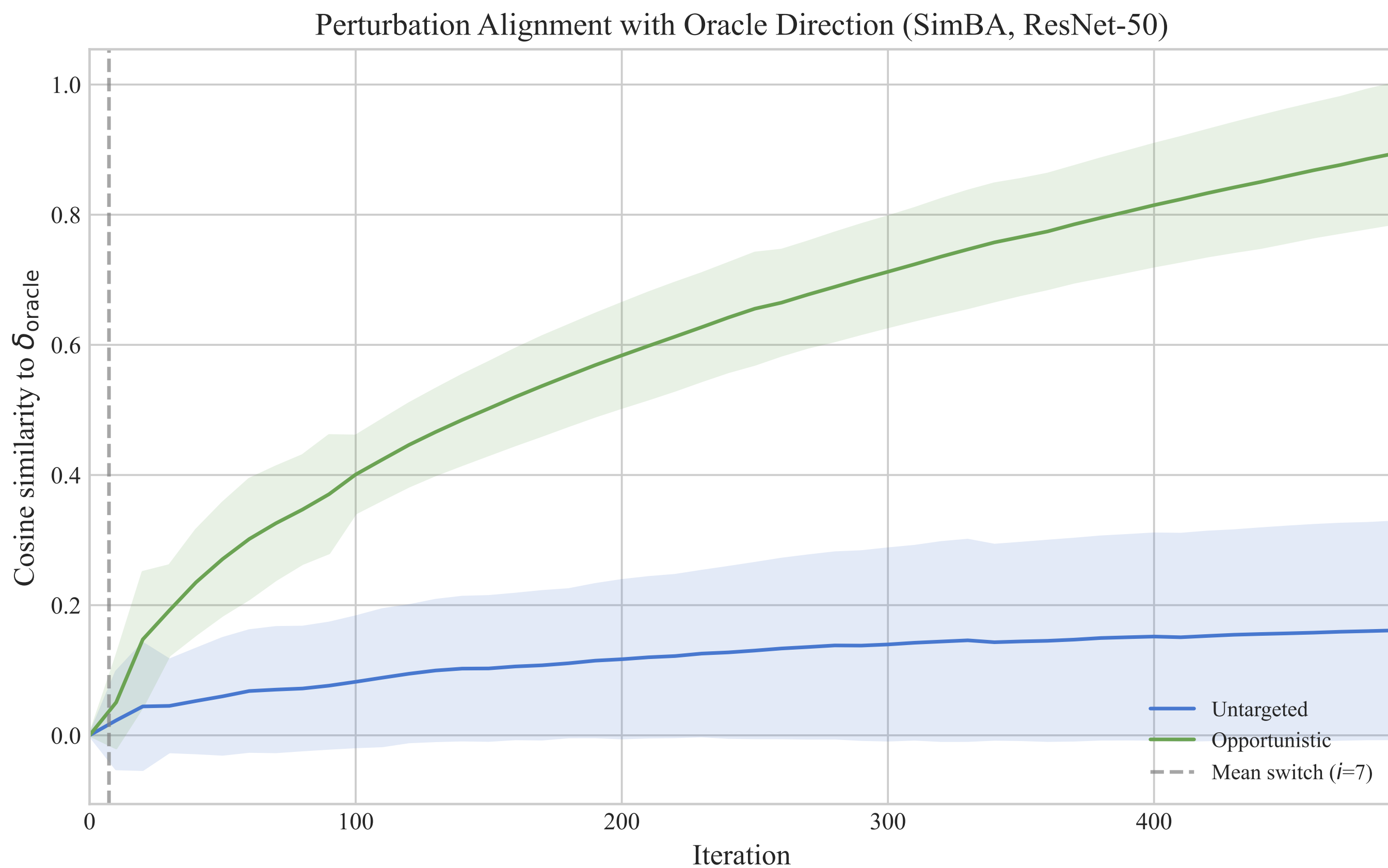
|                |                                                          |
|----------------|----------------------------------------------------------|
| <b>Models</b>  | AlexNet, ResNet-18, VGG-16, ResNet-50, ViT-B/16          |
| <b>Attacks</b> | SimBA, Square Attack with CE loss, Bandits               |
| <b>Modes</b>   | Untargeted baseline, targeted oracle, OTS                |
| <b>Budget</b>  | $\epsilon = 8/255$ in $L_\infty$ , 10K iterations        |
| <b>Scale</b>   | 100 ImageNet images per model, 4,500 standard-model runs |
| <b>Metrics</b> | Success rate and censored mean iterations                |

**Oracle target:** same-seed untargeted endpoint, used as a trajectory-specific ceiling.

### 4. When and Why OTS Helps

| Attack    | Direction source   | Expected OTS effect              |
|-----------|--------------------|----------------------------------|
| SimBA     | No target tracking | Large gain from early commitment |
| Square CE | $y$ -only CE       | Margin-like gain                 |
| Bandits   | Gradient estimate  | Small or negative gain           |

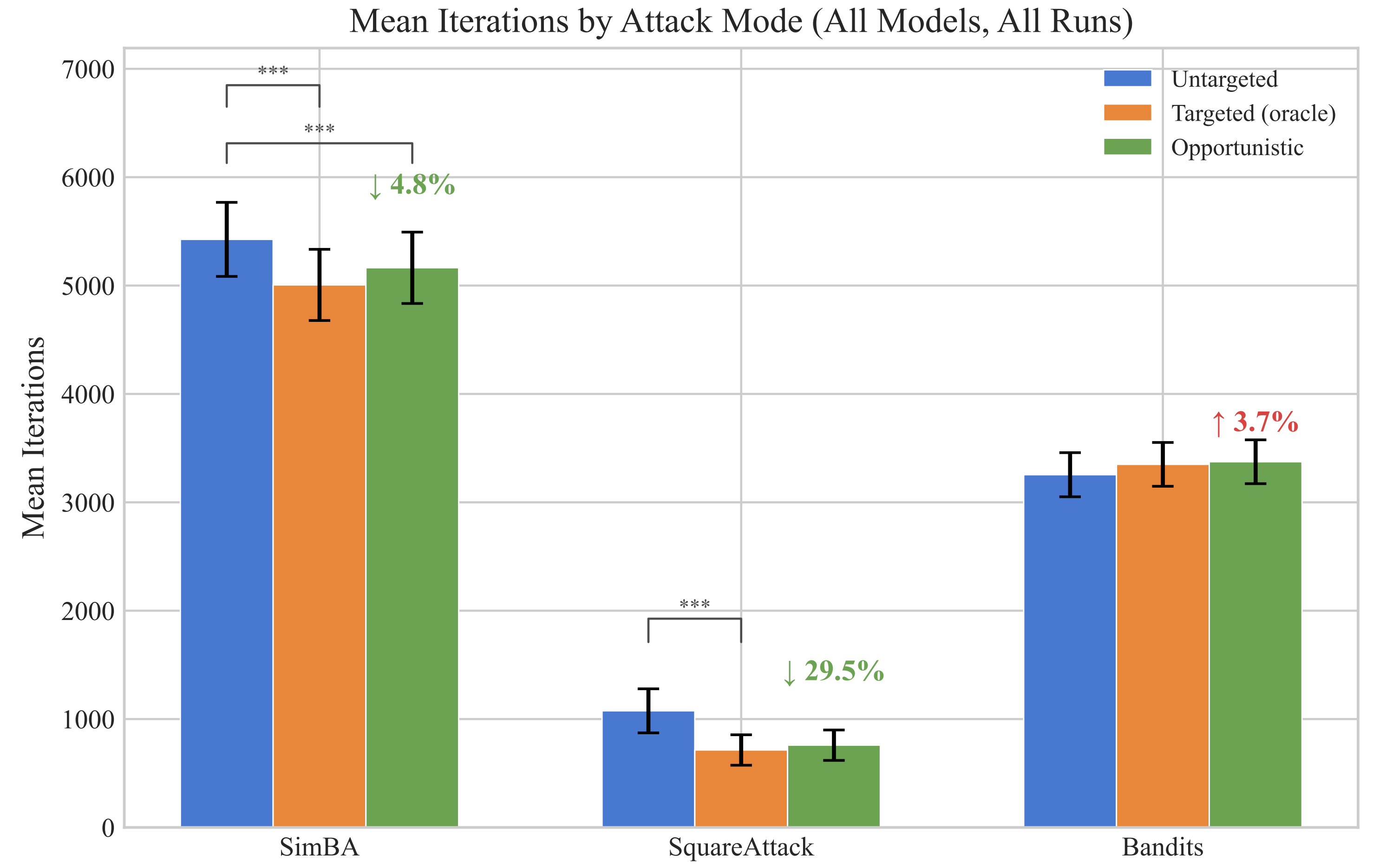
- Helps when the base loss drifts.
- Can hurt when the base attack already has a direction.



- Mechanism: OTS turns early score information into a fixed direction.
- Fixed  $t$  mimics the margin competitor:

$$\mathcal{L}_{\text{margin}} = f_y(x') - \max_{k \neq y} f_k(x').$$

### 5. Main Results



| Attack    | Untargeted | OTS        | Oracle |
|-----------|------------|------------|--------|
| SimBA     | 67%        | <b>75%</b> | 76%    |
| Square CE | 96%        | <b>99%</b> | 99%    |
| Bandits   | 37%        | 34%        | 35%    |

**+27 pp**

SimBA success on ResNet-50

**43%**

relative iteration reduction

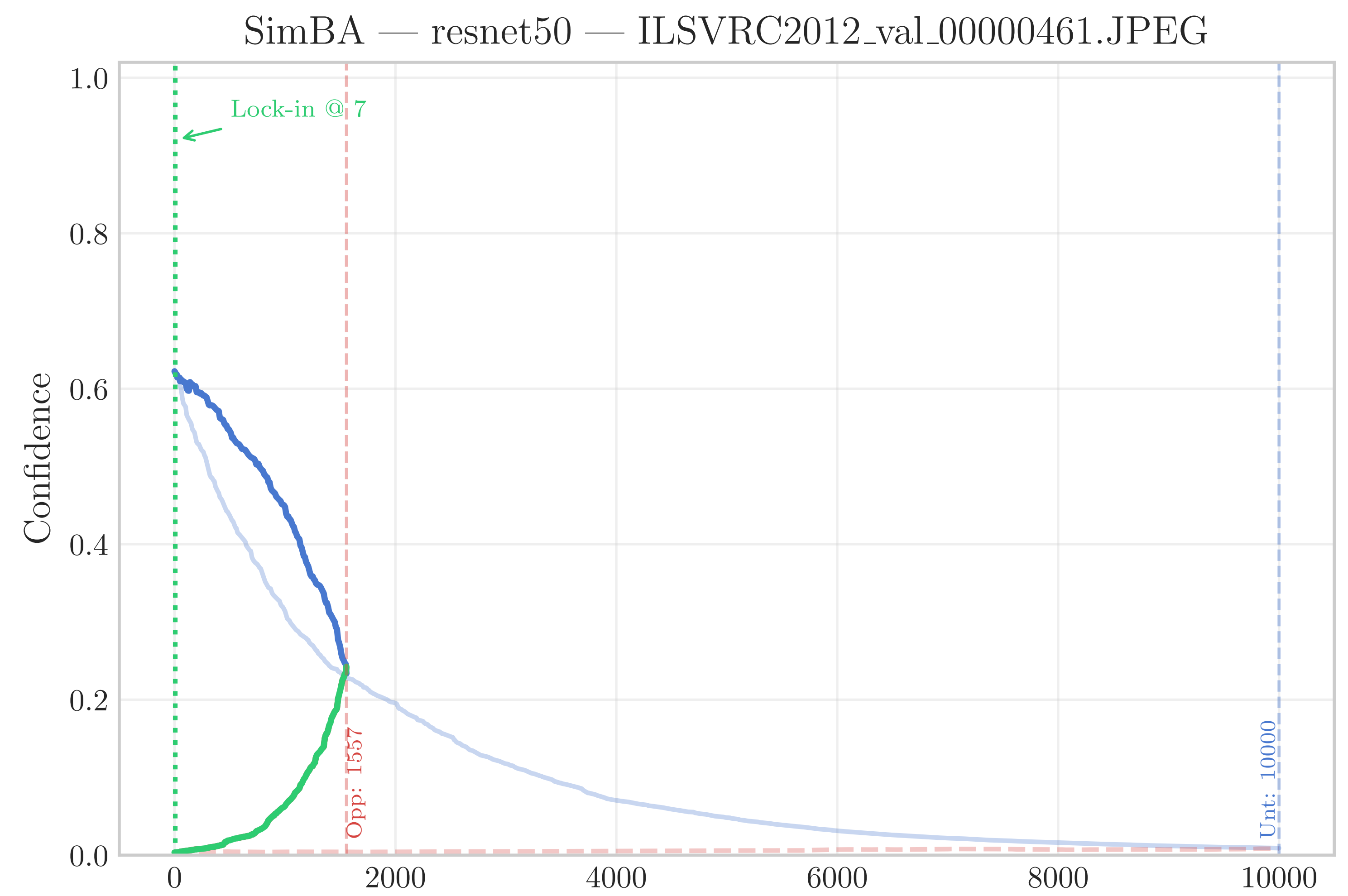
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gradient access

#### Main finding.

- Near-oracle behavior for SimBA and Square CE.
- Bandits gets worse: it already follows an estimated gradient.

### 6. Lock-In Dynamics



- Untargeted: suppresses the true class, then drifts.
- OTS: locks class 988 and keeps increasing it.
- Only a stable competitive target is needed.

### 7. Limits and Takeaways

#### Take-away.

- Explore briefly, then lock the strongest non-true class.
- The gain comes from avoiding class drift under score-only feedback.
- OTS is useful when the base attack lacks a stable direction; it can hurt when the attack already follows one.

#### Current limits.

- Robust ImageNet models: flat success across variants.
- No medium-difficulty regime where commitment pays off.

#### Future work.

- CIFAR-10,  $L_2$  budgets, randomized defenses.
- Adaptive re-locking if the target drops in rank.